

An Integrated Model of Cognition, Emotion, and Vote Choice: Evidence from Taiwan

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Fearful



Angry



Sad



Happy



Disgusted



Surprised

Figure: Ekmanian facial expressions (from Young et al., 2002)



Figure:
<https://cinemasiren.com/cinema-siren/inside-out-review-invention-and-influence-make-for-a-fantastic-flick/>

Introduction & Research Problem

- Traditional voting behavior models focus heavily on cognitive evaluations (retrospective performance, issue stances, candidate assessments) or candidate-centered affect.
- Environment-elicited emotions—feelings triggered by general geopolitical uncertainty, risk, and external coercion—are rarely integrated into models of electoral choice.
- Objective: Develop and empirically test a dual-pathway framework where cognitive evaluations and discrete emotions operate concurrently and interactively.
- Key Question: How do voters combine performance evaluations with environment-elicited anxiety and anger under sustained political pressure?

- Research on voting behavior has long emphasized that citizens rely on cognitive considerations—such as issue preferences, retrospective performance evaluations, and candidate assessments—when making electoral choices (Fiorina 1981; Lau and Sears 1981; Jackson 1975; Kinder and Kiewit 1979; Lodge, Steenbergen, and Brau 1995).
- A complementary tradition demonstrates that affective predispositions, especially partisanship, guide information processing and shape evaluative orientations (Campbell et al. 1960; Jacoby 1988; Greene 1999; Bartels 2000).

- Existing work tends to examine emotions triggered by campaign stimuli or candidate impressions (Abelson et al. 1982; Glaser and Salovey 1998; Isbell and Ottati, 2002). Some research finds how broader political conditions generate emotions and influence attitudes and behavior (Wagner and Morisi 2019; Rico, Guinjoan, and Anduiza 2017; Vasilopoulos et al. 2019), but this work leaves unexplored how these reactions combine with cognitive evaluations to shape electoral decisions.

Theoretical Framework: Dual-Pathway Model

Cognitive Pathway

- Conscious, deliberative performance judgments.
- Retrospective assessments of national standing and diplomatic management.
- Standard electoral accountability mechanism: punishment for perceived decline.

Affective Pathway

- Concurrent discrete emotional reactions grounded in Appraisal Theory and Affective Intelligence Theory (AIT).
- Surveillance System → Anxiety (Uncertainty & Risk).
- Disposition System → Anger (Certainty & Other-Blame).

- **Anxiety (Uncertainty-Based Hazard Appraisal):**

- Heightens vigilance, focuses attention on potential losses, and increases risk aversion.
- Encourages caution and reconsidering status quo under stable conditions.
- In severe crises, risk-aversion may favor familiar, tested leadership over opposition uncertainty.

- **Anger (Blame-Based Appraisal of Intentional Threat):**

- Elicited when threats are certain, intentional, and attributable to an external actor.
- Activates approach motivation, punitive desires, and defensive solidarity.
- Intertwined with incumbent alignment when incumbents pursue deterrence-oriented postures.

Appraisal Theory of Emotion

Cognitive evaluation as the mediator between political stimuli and affective response

1. Primary Appraisal (Goal Relevance)

Evaluates the significance of an event for personal well-being or group identity:

- **Goal Relevance:** Does the event matter?
- **Goal Congruence:** Is it helpful or harmful?
- **Ego-Involvement:** Threat to partisan/national identity.

2. Secondary Appraisal (Coping Options)

Evaluates resources and control mechanisms:

- **Blame/Credit:** Attribution of responsibility.
- **Coping Potential:** Ability to alter or adapt.
- **Future Expectations:** Possibility of outcome shift.

Key Theoretical Insights

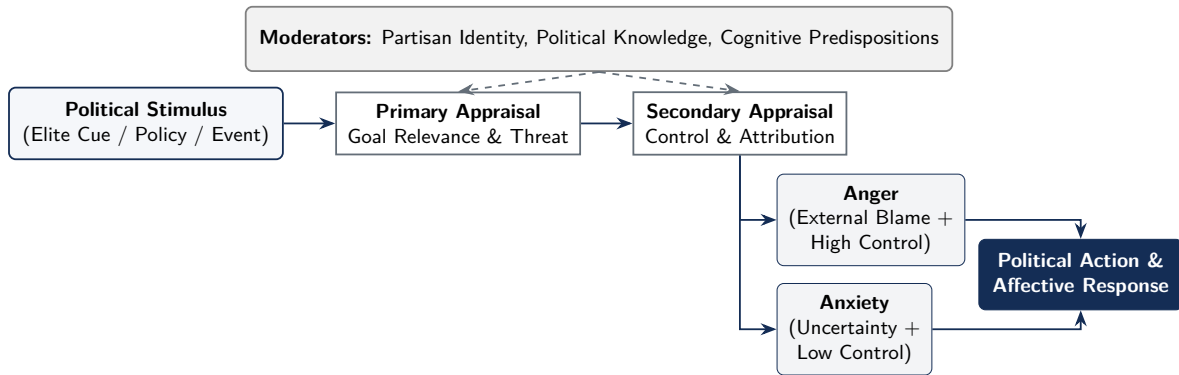
- **Cognitive Primacy:** Emotion is not an unmediated visceral response; it depends on contextual interpretation and elite signaling.
- **Emotion Differentiation:** Identical macro-events elicit distinct emotions based on attribution:
 - High threat + Low control → **Anxiety**
 - High threat + External blame → **Anger**

Action Tendencies in Public Opinion

Anxiety prompts information search and cognitive deliberation; **Anger** energizes partisan mobilization and reinforces motivated reasoning.

Cognitive Appraisal Process Model

Pathways from external stimuli to affective polarization and action



Model Dynamics:

- **Sequential Evaluation:** Stimuli undergo automatic primary checks before secondary attribution processing.

Moderating Role:

- Partisan cues shape both initial threat perception (Primary) and causal attribution (Secondary).

Research Hypotheses

- H1 (Cognitive Punishment): Voters who perceive Taiwan's international or cross-Strait situation as deteriorating will be less likely to support the incumbent party.
- H2 (Anxiety-Induced Weakening): Higher levels of anxiety about potential military conflict will decrease support for the incumbent party.
- H3 (Anger-Induced Strengthening): Higher levels of anger at external coercion will increase support for an incumbent pursuing a deterrence-oriented posture.
- H4 (Anxiety-Contingent Reversal): Under perceptions of severe national decline, anxiety attenuates or reverses the negative effect of that evaluation on incumbent support, as risk-averse voters prefer familiar leadership.

When the perceived threat environment is severe enough, the uncertainty-reduction motive dominates over the scrutiny motive. Under conditions of acute perceived danger, switching to an untested alternative (Hou or Ko) itself becomes a source of risk, so risk-averse, anxious voters gravitate toward the known quantity.

Empirical Context: Taiwan's 2024 Election

- Sustained external military and diplomatic coercion from China made cross-Strait policy and international standing highly salient.
- Incumbent DPP (Lai Ching-te) maintained an assertive deterrence stance and refused negotiation under coercion.
- Opposition parties (KMT Hou Yu-ih, TPP Ko Wen-je) proposed alternative approaches (dialogue/third-way policy).
- Provides a natural setting to observe how cognitive appraisals of national trajectory interact with discrete security emotions.

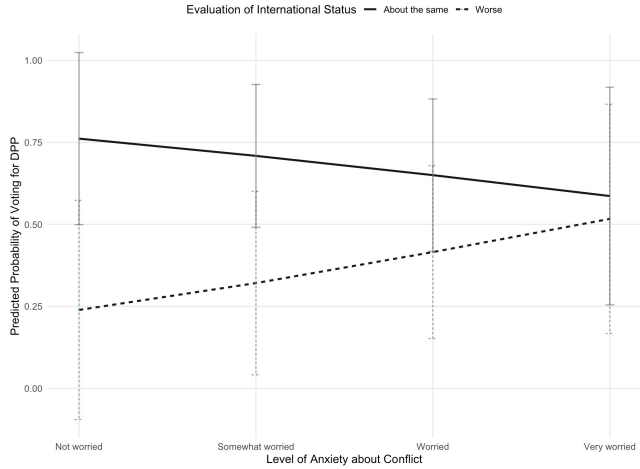
Data and Operationalization

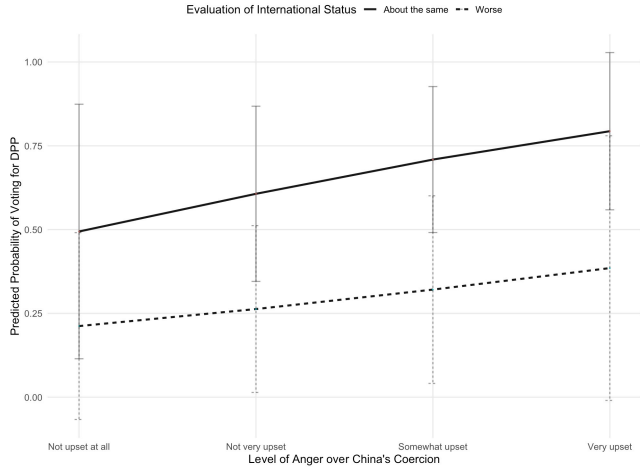
- **Data Source:** Taiwan Election and Democratization Study (TEDS 2024), probability sample ($N = 1,206$; regression $N = 793$).
- **Dependent Variable:** Support for incumbent DPP ticket (binary logit; checked via multinomial models).
- **Key Independent Variables:**
 - Cognitive: 4-year retrospective evaluation of Taiwan's international status and cross-Strait relations.
 - Anxiety: Worry over potential cross-Strait military conflict.
 - Anger: Upset over China's military encirclement/aircraft incursions.
- **Controls:** Lagged 2020 vote choice, Taiwanese identity, sociodemographics, and comparative candidate feeling thermometers (Lai vs. Hou, Lai vs. Ko).

Main Regression Results (Logistic Regression)

Variable	Baseline (1)	Lai-Hou Diff (2)	Lai-Ko Diff (3)	Full Model (4)
International status worse	0.013** (0.019)	0.022* (0.038)	0.035† (0.068)	0.041† (0.079)
Level of Anxiety	0.308** (0.111)	0.333** (0.135)	0.189** (0.107)	0.192** (0.108)
Level of Anger	2.509** (0.838)	2.021† (0.767)	1.718 (0.828)	1.670 (0.810)
Intl Status Worse × Anxiety	3.523** (1.666)	3.010* (1.573)	1.977 (1.244)	1.809 (1.148)
Intl Status Worse × Anger	0.641 (0.285)	0.764 (0.363)	1.055 (0.625)	1.158 (0.687)
Voted DPP in 2020	58.196***	16.453***	31.061***	22.835***
Taiwanese Identity	4.093***	2.940**	2.554*	2.494*
Lai-Hou Feeling Difference	—	1.563***	—	1.201*
Lai-Ko Feeling Difference	—	—	2.233***	2.098***

Table: Odds Ratios shown. Standard errors in parentheses. † $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.





Key Empirical Finding: Asymmetry of Effects

Anxiety Pathway

- Highly stable negative main effect across all specifications ($OR \approx 0.18 - 0.33, p < 0.01$).
- Operates independently of candidate-centered feeling thermometers.
- Reflects a distinct, target-general surveillance mechanism.

Anger Pathway

- Significant positive effect in baseline model ($OR = 2.509, p < 0.01$).
- Attenuates once candidate feeling differences (especially Lai-Ko) are controlled.
- Suggests anger is closely entangled with candidate evaluations and partisan attachments.

Testing H4: Interaction of Anxiety and Evaluation

- Baseline & Partial Controls: The interaction term *International status worse* $\times$ *Anxiety* is positive and statistically significant (OR = 3.523, $p < 0.01$).
- Interpretation:
 - Under stable conditions, anxiety promotes vigilance and punishment (switching from incumbent is perceived as low risk).
 - Under severe perceived decline, anxiety leads voters to consolidate around experienced leadership to avoid the uncertainty of political change.
- Robustness Nuance: The interaction attenuates under full candidate controls, indicating suggestive rather than conclusive evidence for the reversal mechanism.

Summary of Hypotheses Testing

Hypothesis	Theoretical Expectation	Empirical Verdict
H1: Cognitive Punishment	Deterioration reduces support	Supported ($p < 0.05$)
H2: Anxiety Weakening	Anxiety reduces incumbent support	Supported ($p < 0.01$, robust)
H3: Anger Strengthening	Anger increases support for deterrence	Supported in baseline; attenuates under candidate affect controls
H4: Anxiety Reversal	Anxiety reverses punishment under decline	Supported in baseline/partial controls; attenuates in full model

- **Subgroup Analysis (2020 Non-DPP Voters):** Suggestive evidence that anger at external coercion promotes incumbent support beyond core partisans.
- **Multinomial Logit Specification:** Separating KMT and TPP choices confirms that anxiety drives votes away from the DPP to both opposition candidates, while anger reduces opposition support.
- **Survey Weights & Party Controls:** Weighted specifications and models varying party vs. candidate feeling thermometers demonstrate consistent main effects for anxiety.

Academic Contributions & Conclusion

- **Theoretical Refinement:** Demonstrates that cognition and emotion act as concurrent, complementary mechanisms rather than competing explanations in vote choice.
- **Emotional Nuance:** Highlights fundamental asymmetries between anxiety (diffuse hazard, stable independent effect) and anger (target-specific blame, entangled with candidate affect).
- **Contextual Accountability:** Re-evaluates electoral punishment by showing that threat appraisal can alter whether anxiety leads to change or continuity.
- **Broader Implications:** Provides a theoretical framework applicable to other democracies operating under sustained external security coercion.